

Theorem 1 (Menelaus's Theorem). *Let $\triangle ABC$ be a triangle, and let a line l meet the lines BC , CA , and AB (or their extensions) at points D , E , and F , respectively, with D, E, F distinct from A, B, C . Using signed lengths, Menelaus's theorem states that*

$$\frac{\overline{AF}}{\overline{FB}} \cdot \frac{\overline{BD}}{\overline{DC}} \cdot \frac{\overline{CE}}{\overline{EA}} = -1.$$

If one takes absolute values instead, then the corresponding unsigned product is 1.

Proof. An affine transformation preserves collinearity and preserves ratios of directed segments on any fixed line. Therefore it is enough to prove the theorem after sending

$$A = (0, 0), \quad B = (1, 0), \quad C = (0, 1).$$

In these coordinates, the three sideline intersections can be written as

$$F = (f, 0), \quad E = (0, e), \quad D = (d, 1 - d)$$

for some real numbers f, e, d .

Since D, E, F are collinear, their coordinate matrix has determinant zero:

$$\det \begin{pmatrix} f & 0 & 1 \\ 0 & e & 1 \\ d & 1 - d & 1 \end{pmatrix} = 0.$$

Expanding the determinant gives

$$-de + df + ef - f = 0,$$

or equivalently

$$d(e - f) = f(1 - e).$$

Hence

$$d = \frac{f(e - 1)}{e - f}, \quad 1 - d = \frac{e(1 - f)}{e - f}.$$

Now compute the directed ratios along the three sidelines:

$$\frac{\overline{AF}}{\overline{FB}} = \frac{f}{1 - f}, \quad \frac{\overline{BD}}{\overline{DC}} = \frac{1 - d}{d}, \quad \frac{\overline{CE}}{\overline{EA}} = \frac{1 - e}{e}.$$

Therefore

$$\frac{\overline{AF}}{\overline{FB}} \cdot \frac{\overline{BD}}{\overline{DC}} \cdot \frac{\overline{CE}}{\overline{EA}} = \frac{f}{1 - f} \cdot \frac{e(1 - f)}{f(e - 1)} \cdot \frac{1 - e}{e} = -1.$$

This proves Menelaus's theorem. Taking absolute values immediately gives the unsigned version with product 1. $\square$